

SAVOONGA, AK, USA

WMO: 702035

Lat: 63.687N

Lon: 170.493W

Elev: 17

StdP: 101.12

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-27.3	-33.0	0.2	-27.1	-32.3	0.2	-26.1	19.0	-2.2	16.4	-4.7	2.0	200	0.991

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	13.9	10.4	12.5	9.8	11.7	9.3	11.3	13.3	10.3	12.1	9.5	11.3	5.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	7.6	12.3	8.9	7.1	11.3	7.9	6.6	10.1	32.3	13.3	29.6	12.1	27.9	11.0	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.6	14.7	12.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-30.8	13.7	3.4	2.9	-33.2	15.8	-35.2	17.5	-37.1	19.1	-39.6	21.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-14.6	-14.5	-12.9	-8.3	-1.1	4.7	8.4	8.2	5.2	1.1	-3.1	-9.8
	DBStd	9.85	7.58	8.41	6.98	5.88	3.41	2.72	2.31	2.19	2.68	2.89	4.14	7.48
	HDD10.0	4765	762	685	710	551	344	162	61	63	143	277	393	614
	HDD18.3	7787	1021	918	968	801	603	411	308	315	393	535	643	872
	CDD10.0	20	0	0	0	0	0	2	11	7	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	5.6	5.5	5.9	5.1	4.8	4.3	4.2	4.5	5.0	5.7	6.9	8.3	7.6
	PrecAvg	581	42	39	29	26	30	37	60	82	72	62	59	45
(15) Precipitation	PrecMax	887	205	159	80	101	71	85	106	142	123	152	113	103
	PrecMin	347	1	5	0	4	10	8	18	16	29	13	10	1
	PrecStd	123	40	32	23	19	18	16	21	26	25	31	32	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	2.1	2.1	2.5	7.7	17.3	17.9	14.8	12.2	9.1	5.0	2.4
		MCWB	0.6	0.3	0.7	0.8	5.2	11.8	12.4	11.2	10.2	7.9	2.0	0.8
	2%	DB	1.0	1.0	1.0	1.4	4.1	12.6	14.9	12.9	11.1	7.7	3.0	1.4
		MCWB	-0.2	-0.1	-0.2	0.0	2.1	8.9	11.1	9.9	9.0	6.5	1.6	0.3
	5%	DB	-1.0	-0.1	-0.4	1.0	2.9	10.2	12.9	12.3	9.9	6.3	2.6	0.8
		MCWB	-1.5	-0.7	-1.1	-0.2	1.2	7.5	10.1	9.7	8.2	4.7	1.3	-0.2
	10%	DB	-2.6	-1.4	-2.1	-0.1	2.4	8.3	12.1	11.3	9.0	5.0	2.1	-1.1
		MCWB	-3.5	-1.9	-2.9	-1.0	0.9	6.3	9.7	9.1	7.6	3.6	0.9	-1.6
	0.4%	WB	0.7	0.7	0.8	1.2	5.2	12.1	13.5	12.3	10.3	8.0	3.0	1.2
		MCDB	1.5	1.4	1.4	2.2	7.7	17.1	16.9	13.9	11.6	8.7	3.6	1.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-0.2	0.0	-0.2	0.2	2.8	9.1	11.7	10.9	9.1	6.8	2.3	0.4
		MCDB	0.6	0.7	0.6	1.2	4.3	12.2	14.0	12.2	10.3	7.7	3.2	1.2
	5%	WB	-1.6	-0.8	-1.2	-0.2	1.7	7.7	10.5	10.2	8.3	5.2	1.4	-0.3
		MCDB	-0.9	-0.1	-0.5	0.7	3.0	9.8	12.5	11.6	9.4	6.6	2.5	0.5
	10%	WB	-3.5	-2.1	-2.7	-1.2	0.9	6.5	9.7	9.4	7.8	3.8	0.7	-1.7
		MCDB	-2.6	-1.6	-2.1	-0.4	2.1	8.2	11.7	10.9	8.8	5.1	1.8	-1.2
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.0	6.1	5.2	3.2	4.6	4.5	3.7	3.2	2.8	3.3	4.4
		MCDBR	6.6	5.9	5.8	4.2	4.2	8.5	7.2	5.2	4.2	3.6	3.4	4.6
		MCWBR	6.2	5.6	5.4	3.6	3.5	5.8	4.9	3.9	3.2	3.2	3.2	4.2
	5% WB	MCDBR	6.5	5.8	5.6	4.2	4.3	7.7	6.8	4.6	3.8	3.6	3.2	4.7
		MCWBR	6.2	5.6	5.3	3.7	3.6	5.5	4.8	3.6	3.1	3.3	3.2	4.5
(40) Clear-Sky Solar Irradiance	taub	0.186	0.270	0.280	0.342	0.375	0.374	0.379	0.399	0.338	0.307	0.408	N/A	
	taud	1.956	2.065	2.154	2.020	2.169	2.384	2.424	2.297	2.441	2.269	2.689	N/A	
	Ebn at Noon	427	636	810	820	830	844	829	765	753	596	314	N/A	
	Edh at Noon	29	67	92	133	127	105	99	102	72	55	29	N/A	
(44) All-Sky Solar Radiation	RadAvg	0.17	0.85	2.53	4.10	4.44	5.32	4.24	3.10	2.00	0.84	0.19	0.06	
	RadStd	0.02	0.12	0.25	0.46	0.45	0.37	0.37	0.33	0.17	0.10	0.02	0.01	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page